

# Manual for running MARS-\* series

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# 1 General remarks

The MARS-\* series of codes include MARS-F [3], MARS-K [4] and MARS-Q [8] so far. These are various extensions to the MARS code [1, 2] that solves the single fluid, ideal or resistive MHD equations with toroidal flow. MARS-F includes the capability of feedback control of MHD instability (e.g. the RWM). MARS-K is a MHD-drift kinetic hybrid implementation. MARS-Q is a quasi-linear expansion. The equilibrium input to MARS-\* is mainly produced by the CHEASE code [5], which is a fixed boundary Grad-Shafranov equilibrium solver.

This manual intends to give some guidance on how to run the MARS-\* series to study various physics problems associated with macroscopic MHD instabilities as well as macroscopic plasma response to external perturbations. This by no means covers the full aspects of the usage of the code. Computational skills and physics insights are essential in obtaining correct numerical results. The wisdom of "anything that can go wrong will go wrong" happily applies when running the codes.

The following sections aim at a reasonably systematic description of how to run the codes. The readers can refer to the FAQ Section towards the end, for some quick answers.

Throughout this manual, we adopt certain colouring convention to denote the **Namelist input parameters for CHEASE**, the **Namelist input parameters for MARS-\***, the **input data files for MARS-\***, as well as the **output data files from MARS-\***.

## 2 Some basic elements in MARS-\*

### 2.1 Equilibrium magnetic field

The equilibrium magnetic field is represented as

$$\mathbf{B} = \nabla\phi \times \nabla\psi + T(\psi)\nabla\phi \quad (1)$$

where  $\phi$  is the geometrical toroidal angle,  $\psi$  the equilibrium poloidal magnetic flux,  $T$  the poloidal current flux function. Note that this representation differs from a form  $\mathbf{B} = \nabla\psi \times \nabla\phi + T(\psi)\nabla\phi$  that is also often adopted by others.

The positive toroidal angle  $\phi$  is defined in the *clockwise* direction when viewed from the positive  $Z$ , effectively forming a left-handed cylindrical coordinate system  $(R, Z, \phi)$ .

### 2.2 Curve-linear coordinate system

MARS-\* solves the MHD-kinetic equations in a magnetic flux surface based curve-linear coordinate system  $(s, \chi, \phi)$ , where the radial coordinate  $s \equiv \sqrt{(\psi - \psi_{axis})/(\psi_{edge} - \psi_{axis})}$  labels the equilibrium flux surface,  $\chi$  is a generalised poloidal angle whose definition depends on the choice of the Jacobian (see Section 2.3). Note that this is generally *not* an orthogonal coordinate system - the radial and the poloidal coordinates are not orthogonal to each other, though both are orthogonal to the toroidal coordinate. Associated with this coordinate system are the contra-variant basis vectors

$$\mathbf{e}^s = \nabla s, \quad \mathbf{e}^\chi = \nabla \chi, \quad \mathbf{e}^\phi = \nabla \phi, \quad (2)$$

the covariant basis vectors

$$\mathbf{e}_s = \partial_s \mathbf{r} = J \nabla \chi \times \nabla \phi, \quad \mathbf{e}_\chi = \partial_\chi \mathbf{r} = J \nabla \phi \times \nabla s, \quad \mathbf{e}_\phi = \partial_\phi \mathbf{r} = J \nabla s \times \nabla \chi = R^2 \nabla \phi, \quad (3)$$

the Jacobian of the coordinate transform

$$J = [\mathbf{e}_s \cdot \mathbf{e}_\chi \times \mathbf{e}_\phi] = [\nabla s \cdot \nabla \chi \times \nabla \phi]^{-1}, \quad (4)$$

as well as the metrics elements

$$g_{ss} = \mathbf{e}_s \cdot \mathbf{e}_s, \quad g_{s\chi} = \mathbf{e}_s \cdot \mathbf{e}_\chi, \quad g_{\chi\chi} = \mathbf{e}_\chi \cdot \mathbf{e}_\chi, \quad g_{\phi\phi} = \mathbf{e}_\phi \cdot \mathbf{e}_\phi, \quad (5)$$

$$g^{ss} = \mathbf{e}^s \cdot \mathbf{e}^s, \quad g^{s\chi} = \mathbf{e}^s \cdot \mathbf{e}^\chi, \quad g^{\chi\chi} = \mathbf{e}^\chi \cdot \mathbf{e}^\chi, \quad g^{\phi\phi} = \mathbf{e}^\phi \cdot \mathbf{e}^\phi. \quad (6)$$

The Jacobian is also calculated as

$$J = \sqrt{g_{\phi\phi}(g_{ss}g_{\chi\chi} - g_{s\chi}^2)}. \quad (7)$$

## 2.3 Choice of the magnetic coordinates

Various magnetic coordinates - essentially the poloidal angle  $\chi$  - are defined by choosing a Jacobian of the form

$$J = C(\psi)R^\alpha|\nabla\psi|^\mu, \quad (8)$$

where  $\alpha = \text{NER}$  and  $\mu = \text{NEGP}$ . In particular, the choice of  $(\alpha = 2, \mu = 0)$  gives the PEST-like straight field line (SFL) coordinates; the choice of  $(\alpha = 1, \mu = -1)$  gives the equal-arc coordinates which is not a SFL system.

The SFL coordinates are often numerically more accurate for tearing mode computations, whilst the equal-arc coordinates have better numerical convergence properties for computing pressure driven kink mode or the resistive wall mode (RWM) which exhibits ballooning characteristics in the low field side (LFS) of the torus.

Note that the specific form (8) of the choice of the Jacobian excludes the Boozer coordinates as well as the Hamada coordinates, although if needed, it is possible to *a posteriori* map the MARS-\* results into the aforementioned coordinates. [A specific choice of  $(\alpha = 0, \mu = 0)$  is allowed, which, in combination of rescaling the radial coordinate, can give  $J = 1$ . But this is neither the Hamada nor a SFL system.]

## 2.4 Perturbed quantities

The essential part of the equations that MARS-\* solve is the perturbed resistive MHD equation. The simplest version (without equilibrium flow and other physics effects) is shown below

$$\gamma\rho\mathbf{v} = \mathbf{j} \times \mathbf{B} + \mathbf{J} \times \mathbf{b} - \nabla p, \quad (9)$$

$$\gamma\mathbf{b} = \nabla \times (\mathbf{v} \times \mathbf{B}) - \nabla \times (\eta\mathbf{j}), \quad (10)$$

$$\mathbf{j} = \nabla \times \mathbf{b}, \quad (11)$$

$$\gamma p = -\Gamma P \nabla \cdot \mathbf{v} - \mathbf{v} \cdot \nabla P, \quad (12)$$

where  $\mathbf{B}, \mathbf{J}, P, \rho$  are equilibrium quantities, and  $\mathbf{b}, \mathbf{j}, \mathbf{v}, p$  are perturbed quantities.

The perturbed vector quantities are decomposed as

$$\mathbf{v} = v^1\mathbf{a}_1 + v^2\mathbf{a}_2 + v^3\mathbf{a}_3, \quad (13)$$

$$\mathbf{b} = b^1\nabla\chi \times \nabla\phi + b^2\nabla\phi \times \nabla s + b^3\nabla s \times \nabla\chi, \quad (14)$$

$$\mathbf{j} = j^1\nabla\chi \times \nabla\phi + j^2\nabla\phi \times \nabla s + j^3\nabla s \times \nabla\chi, \quad (15)$$

where  $v^{1,2,3}, b^{1,2,3}, j^{1,2,3}$  are (part of) the solution variables, and

$$\mathbf{a}_1 = \left( \frac{\mathbf{B}}{B} \times \mathbf{e}_s \right) \times \frac{\mathbf{B}}{B}, \quad \mathbf{a}_2 = J \frac{\mathbf{B} \times \mathbf{e}^s}{B^2}, \quad \mathbf{a}_3 = \mathbf{B}. \quad (16)$$

Table 1: Normalisation factors in MARS-<sup>\*</sup>.

| quantity        | SI unit           | normalisation factor           |
|-----------------|-------------------|--------------------------------|
| length          | m                 | $R_0$                          |
| mag.field       | Tesla             | $B_0$                          |
| density         | kg/m <sup>3</sup> | $\rho_0$                       |
| speed           | m/s               | $v_A = B_0/\sqrt{\mu_0\rho_0}$ |
| time            | s                 | $\tau_A = R_0/v_A$             |
| frequency       | rad/s             | $\omega_A = 1/\tau_A$          |
| pressure        | Pascal            | $B_0^2/\mu_0$                  |
| torque density  | N/m <sup>2</sup>  | $B_0^2/\mu_0$                  |
| torque          | N·m               | $R_0^3 B_0^2/\mu_0$            |
| current density | A/m <sup>2</sup>  | $B_0/(\mu_0 R_0)$              |
| current         | A                 | $R_0 B_0/\mu_0$                |
| resistivity     | Ohm·m             | $\mu_0 R_0 v_A/A^2$            |

## 2.5 Conversion of MARS-<sup>\*</sup> quantities into cylindrical coordinates

We give an example for the perturbed magnetic field. Knowing the MARS-<sup>\*</sup> computed  $b^{1,2,3}$ , we can compute the cylindrical components as

$$b_R = \frac{1}{J} \left( \frac{\partial R}{\partial s} b^1 + \frac{\partial R}{\partial \chi} b^2 \right), \quad b_Z = \frac{1}{J} \left( \frac{\partial Z}{\partial s} b^1 + \frac{\partial Z}{\partial \chi} b^2 \right), \quad b_\phi = \frac{R}{J} b^3. \quad (17)$$

Furthermore, the normal component of the **b**-field is

$$b_n = \frac{\mathbf{b} \cdot \nabla s}{|\nabla s|} = \frac{b^1}{J|\nabla s|} = \frac{b^1}{R\sqrt{g_{\chi\chi}}}. \quad (18)$$

## 2.6 Normalization

The basic normalisation factors are  $R_0 \equiv \text{R0EXP}$  for the length,  $B_0 \equiv \text{B0EXP}$  for the magnetic field,  $\rho_0$  for the plasma density, where  $R_0$  is not uniquely defined but often chosen as the major radius of either the machine centre or the magnetic axis,  $B_0$  is the vacuum toroidal field at  $R_0$ ,  $\rho_0$  is the plasma density at the magnetic axis. The derived normalisation factors, together with the above basic ones, are summarised in Table 1. Both equilibrium and perturbed quantities are subject to the same normalisation.

The factor  $A \equiv R_0/a$  in the normalisation factor for the plasma resistivity is the aspect ratio of the plasma. Note that the plasma resistivity is normalised in such a way, that the dimensionless resistivity in MARS-<sup>\*</sup> input [ETA](#) is exactly equal to the inverse of the Lundquist number  $S \equiv \tau_R/\tau_A$  with  $\tau_R \equiv \mu_0 a^2/\eta$ . Also note that the Alfvén frequency  $\omega_A$  is not  $2\pi/\tau_A$ , due to the absence of the  $2\pi$ -factor in the definition of the Alfvén time  $\tau_A$ .

## 2.7 Comments on the resistive wall time in MARS-\*

In MARS-\*, the resistivity  $\eta_w^M$  of the wall is specified via an input parameter  $\tau_w^M = \text{TAUW}$

$$\eta_w^M = \frac{(s_w/a)(d_w/a)}{\tau_w^M A^2}, \quad (19)$$

where  $s_w$  is the minor radius of the wall (specified in the  $s$ -coordinate),  $d_w$  the radial cell size occupied by the (thin) wall,  $A \equiv R_0/a$  the aspect ratio. Note that the input parameter  $\tau_w^M$  here does not specify the true wall time, but rather serves as an input quantity to help specify the wall resistivity according to the above Eq. (19). This fictitious wall time  $\tau_w^M$  is normally a reasonable guess (by the order of magnitude) for the true wall time  $\tau_w^C$ , which can be computed by running MARS-\* in the “vacuum mode”, i.e. by replacing all the plasma (MHD) equations by the vacuum equations, and solve a wall eddy current decay eigenvalue problem, with the given wall resistivity  $\tau_w^M$  and the true wall geometry. Normally we define the wall time as that corresponding to the slowest decaying wall eigenmode, for a given toroidal mode number  $n$ .

For the RWM, the mode’s eigenvalue is often normalised by the wall time. The best choice is the MARS-\* computed wall time  $\tau_w^C$ . However, since  $\tau_w^C$  scales linearly with  $\tau_w^M$ , and since both numbers are in the same order of magnitude, we sometimes also normalise the RWM eigenvalue simply by the input parameter  $\tau_w^M$ .

In MARS-\*, the wall shape does not have to be conformal to the plasma boundary. It is possible to specify multiple walls. It is also possible to specify resistive walls with varying wall resistivity along the poloidal circumference. An extreme case of the latter possibility is to specify a partial wall that has vacuum gap(s) along the poloidal angle. However, in all cases, the wall is assumed uniform and continuous along the toroidal angle. Refer to Sec. 3.2 and Sec. 4.2.1 for detailed settings.



## 3 Equilibrium computation using CHEASE

### 3.1 A short introduction

The equilibrium input for MARS-\* is computed by the CHEASE code. Details of the CHEASE solver are documented in Ref. [5]. This manual only describes key elements of practical importance, that are directly relevant for running MARS-\*, as well as some new features that are not contained in [5].

CHEASE solves a fixed boundary, static, 2D equilibrium problem (the Grad-Shafranov equation), usually for the tokamak configuration, though an RFP option is also available with the new version of the code. The typical input data to CHEASE are: (i) the EXPEQ data (Sec. 3.2) or the EQDSK data (g-file, Sec. 3.3), and (ii) the Namelist input file “datain” (Sec. 3.4). The output data from CHEASE, that serve as the input to MARS-\*, are **OUTRMAR** and **OUTVMAR**, with the former containing the plasma equilibrium data and the latter containing the vacuum data (when applicable).

CHEASE and MARS-\* share the same normalisation convention.

### 3.2 Format for EXPEQ file

The data format for the EXPEQ file, described below, assumes that the equilibrium is fully *numerically* specified, which is the most common case. It is possible to replace part of this file, or the whole of it, by analytic specifications via the Namelist input “datain”.

```
-----  
a/R0  
Z0/R0  
Pedge  
  
NBPS NWBPS NDATA  
R Z (plasma surface)  
R Z [sigma] (wall 1)  
...  
R Z [sigma] (wall NWBPS-1)  
  
NPPF1 NSTTP  
FCSM  
RPPF  
RFUN  
-----
```

The first three numbers specify the inverse aspect ratio, the initial guess for the (normalised) vertical position of the magnetic axis, and the equilibrium plasma edge pressure (normally assuming zero), respectively.

Table 2: Specification of RFUN in the EXPEQ data for CHEASE.

| NSTTP | RFUN=                                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------------|
| 1     | $TT' = T(dT/d\psi)$                                                                                                                |
| 2     | surface averaged toroidal current density $\oint J_\phi(J/R)d\chi / \oint (J/R)d\chi$                                              |
| 3     | surface averaged parallel current density $\oint \mathbf{J} \cdot \mathbf{B} J d\chi / \oint \mathbf{B} \cdot \nabla \phi J d\chi$ |

NBPS = the number of the (R,Z) data points, normalised by  $R_0$ , used to specify the plasma boundary shape as well as the wall shape(s). NWBPS = the total numbers of surfaces to be specified, including the plasma surface (the first set of (R,Z) data) and the wall surface(s). For example, NWBPS=1 if only the plasma surface is to be specified, NWBPS=2 if the plasma surface and one wall surface are to be specified, and so on. All the surfaces should have the same number (NBPS) of the (R,Z) data points. Generally there is no need to specify the wall surface(s) if only conformal, complete wall(s) will be considered in further MARS-\* runs.

To specify a wall with varying resistivity along the poloidal circumference, set NDATA=3, and add a third column after the (R,Z) data, specifying wall conductivity indicator [sigma], for all the wall shapes (no need to do so for the plasma boundary data). The value of [sigma] varies between 0 and 1, with 0 indicating vacuum. For the case of complete wall(s) with uniform resistivity, set NDATA=2, and no need to specify the [sigma] data.

NPPF1 = the dimension of the following three 1D arrays FCSM, RPPF and RFUN. FCSM is the radial  $s$ -grid varying from 0 (magnetic axis) to 1 (the plasma surface). RPPF =  $dP/d\psi$  is the usual RHS term for the Grad-Shafranov equation, defined on the FCSM grid. The meaning of RFUN, also defined on the FCSM grid, depends on the specifier NSTTP, according to Table 2. CHEASE cannot specify the  $q$ -profile as the input data.

### 3.3 Read EQDSK data instead of EXPEQ

CHEASE reads in the EQDSK data by setting the Namelist parameter **NEQDSK**=1 (the normal option). By default (**NEQDSK**=0), CHEASE reads in the EXPEQ data as described in Sec. 3.2. The name of the input file for both the EQDSK data and the EXPEQ data is the same, aka “EXPEQ”.

After reading the data, CHEASE prints out some key equilibrium information directly from EQDSK, which are useful for future re-normalisation of the MARS-\* results or simply for double checking of the equilibrium. For instance, the plasma major radius **R0EXP**, the toroidal field amplitude **B0EXP**, the total plasma current **I0EXP**, the  $q$ -profile, etc.. These key information can be found in the CHEASE log-file by searching for the key word “EQDSK”.

CHEASE can also write out the computed (fixed boundary) equilibrium in the EQDSK data (output file “EQDSK.OUT”), by setting the input Namelist parameter **NIDEAL**=6.

With **NIDEAL**=0 (the option for CHEASE to produce input data for MARS-\*), CHEASE produces an output file “EXPEQ.OUT” in the EXPEQ data format as described in Sec. 3.2. For the purpose of further

modification/refining the equilibrium, it is practically always useful to save this file, and modify/use it as the new input data for further CHEASE runs.

### 3.4 Namelist input for CHEASE

Tables 3- 4 list some key input parameters as well as typical options for CHEASE, as defined in Namelist EQDATA. In some cases, the type (I=integer, R=real, C=complex) of the parameter is indicated. Note that this is only part of the Namelist that is most commonly used. The full list contains 156 parameters in the present version. Also not all the options for each parameters are listed. The meaning as well as the range of values for certain parameters have been extended compared to the original CHEASE version [5].

### 3.5 Smoothing the plasma boundary

Note that CHEASE-MARS-\* cannot deal with a divertor plasma configuration with exact X-point(s). Therefore, after reading in (e.g. from the g-file) the experimental plasma boundary shape for a divertor configuration, it is often necessary to check and, if necessary, to remove the separatrix. There are several ways to accomplish this. One way is to cut away a small portion of flux surfaces near the plasma edge. This can be achieved in CHEASE by shaving the  $q$ -profile near the plasma edge. The procedure is as follows.

1. Set target  $q_{\text{edge}}$  value by setting the Namelist variable **QSHAVE**= $q_{\text{edge}}$ .
2. Run CHEASE.
3. Use the output file EXPEQ.OUT as the new input EXPEQ file, and run CHEASE again.
4. Use option **NCSCAL=2** if the total plasma current is to be kept the same as the un-shaved equilibrium.
5. Adjust plasma pressure (i.e. change Namelist variable **CFBAL**) if  $\beta_N$  is to be kept the same.  $q_{\text{edge}}$  can be slightly modified after this step.
6. Run CHEASE the 3rd time with the new **CFBAL** and with a larger **QSHAVE** value (e.g. increased by 1), and save the output file EXPEQ.OUT as the new EXPEQ with shaved  $q$ .

Note that the Namelist variable **QSHAVE** does not have immediate effect on the equilibrium output data **OUTRMAR** and **OUTVMAR** after each CHEASE run - it merely yields an output file EXPEQ.OUT which specifies a new equilibrium. Also, setting a very large value of **QSHAVE** dis-enables this functionality. Finally, the above procedure may need to be repeated several times to reach the desired result.

The drawback of the above procedure is that after each cutting, all the equilibrium profiles are re-scaled so that the new plasma boundary remains  $s = 1$ . Moreover, either the total plasma current or the total beta is not well conserved.

Table 3: Key input parameters and typical options for CHEASE Namelist EQDATA: I

| parameter | values | description                                                                                    |
|-----------|--------|------------------------------------------------------------------------------------------------|
| B0EXP     | R      | vacuum toroidal field [T] at R0EXP                                                             |
| CFBAL     | R      | scaling factor for equilibrium pressure                                                        |
| EPSLON    | R      | accuracy tolerance of non-linear iteration for G-S solver                                      |
| NCHI      | I      | poloidal resolution of final flux mesh for MARS-*                                              |
| NCSCAL    | 1      | scale equilibrium by specifying $q=QSPEC$ at radius $s=CSSPEC$                                 |
|           | 2      | scale equilibrium by specifying total (normalised) current CURRT                               |
|           | 4      | no scaling of the equilibrium                                                                  |
| NEGP      | I      | power factor for defining jacobian                                                             |
| NEQDSK    | 0      | read in EXPEQ data (NSURF=6 or NPPFUN=4, NFUNC=4)                                              |
|           | 1      | read in standard EQDSK data (g-file)                                                           |
| NER       | I      | power factor for defining jacobian                                                             |
| NMESHA    | 0      | no packing of final radial mesh for MARS-*                                                     |
|           | 1      | pack final radial mesh at $s=APLACE$ with width AWIDTH                                         |
|           | 2      | pack final radial mesh at $q=QPLACE$ with width QWIDTH                                         |
|           | 3      | automatic packing of final radial mesh at all rational surfaces                                |
|           | 4      | automatic packing at all flow induced Alfvén continuum resonance surfaces                      |
|           | 5      | as 3, but also pack final mesh near plasma boundary                                            |
| NMESHA    | 6      | as 4, but also pack final mesh near plasma boundary                                            |
| NPOINDQ   | I      | total number of packed surfaces for manual radial mesh packing options                         |
| NPSI      | I      | radial resolution of final flux mesh for MARS-*                                                |
| NQMIN     | 0      | no scaling of $q_{\min}$                                                                       |
|           | 1      | scale equilibrium with $q_{\min}=QSPEC$                                                        |
| NRFP      | 0      | tokamak equilibrium solver                                                                     |
|           | 1      | RFP equilibrium solver, with pressure and current profiles analytically specified              |
| NS        | I      | radial resolution of the (refined) polar mesh for solving the G-S equation                     |
| NSURF     | < 6    | analytic specification of plasma boundary                                                      |
|           | 6      | numerical specification of plasma boundary via EXPEQ or EQDSK                                  |
| NT        | I      | poloidal resolution of the (refined) polar mesh for solving the G-S equation                   |
| NTOR      | I      | (positive) toroidal mode number used for automatic radial mesh packing                         |
| NV        | I      | radial resolution of final vacuum mesh for MARS-*                                              |
| NVEXP     | 0      | equidistant vacuum s-mesh                                                                      |
|           | 1      | as 0, but mesh points close to walls are shifted to match wall positions                       |
|           | 2      | as 1, with first grid point in vacuum shifted to match the last cell size from the plasma side |
|           | 3      | exponential mesh without considering wall positions                                            |
|           | 4      | exponential mesh with matched wall positions and shifted first vacuum grid                     |
|           | 5      | as 1, with first two cells size half of the remaining cells size                               |
|           | 6      | as 2, with first two cells size half of the remaining cells size                               |
|           | 8      | as 4, with exponential mesh up to first wall and equidistant mesh outside it                   |
|           | 10     | as 1, but specifically for ITER double wall with uniform wall thickness                        |
|           | 20     | as 2, but specifically for ITER double wall with uniform wall thickness                        |
|           | 50     | as 10, with first two cells size half of the remaining cells size                              |
|           | 60     | as 20, with first two cells size half of the remaining cells size                              |
|           | 80     | as 8, but specifically for ITER double wall with uniform wall thickness                        |
|           | 800    | as 80, but with one more surface for ITER RMP coils                                            |

Table 4: Key input parameters and typical options for CHEASE Namelist EQDATA: II

| parameter     | values | description                                                                               |
|---------------|--------|-------------------------------------------------------------------------------------------|
| <b>QSHAVE</b> | R      | shave $q$ -profile near plasma edge, see Sec. 3.5 for more details                        |
| <b>REXT</b>   | R      | minor radius of computational boundary, normalised by $a$                                 |
| <b>R0EXP</b>  | R      | major radius [m] of the torus                                                             |
| <b>ROTE</b>   | R      | (normalised) toroidal rotation amplitude used for automatic radial mesh packing with flow |
| <b>R0W</b>    | R      | origin (R) used for linear scaling of vacuum mesh up to <b>REXT</b>                       |
| <b>RZ0W</b>   | R      | origin (Z) used for linear scaling of vacuum mesh up to <b>REXT</b>                       |
| <b>SOLPDA</b> | R      | fraction of radial grid points used for “unpacked” region                                 |

A less automatic but preferable way is to directly and locally smooth the plasma boundary near the X-point. This has to be carried out manually, but normally resulting in better conservation of the original equilibrium quantities (e.g. the total plasma current) and radial profiles, except the  $q$ -profile near the plasma edge, which we often wish to reduce in order to improve numerical accuracy of the MARS-\* computations.

### 3.6 How to perform pressure scan?

For the purpose of computing e.g. the beta limits due to external ideal kink instability, one needs to scan the plasma pressure based on a given equilibrium. There are more than one way to perform the pressure scan with CHEASE, but the straightforward way is to change the value of the Namelist parameter **CFBAL** with the present version of the code.

The self-consistency of the equilibrium means that it’s difficult to keep the other equilibrium quantities unchanged while scanning the plasma pressure. From the MHD stability point of view, we wish to keep the safety factor (both amplitude and radial profile) as un-affected as possible.

One way to achieve this is to fix the total plasma current by specifying **NCSCAL**=2 and **CURRT**. The other way is to specify the  $q$ -value at one radial location. For example, fix  $q_{\min}$  by setting **NQMIN**=1 and **QSPEC**= $q_{\min}$ . The former way often tends to better conserve the whole radial shape of the  $q$ -profile, while yielding slight variation of the  $q$ -values everywhere across the plasma column.

Table 5: Input Namelists for MARS-.\*.

|          |                                                                               |
|----------|-------------------------------------------------------------------------------|
| BASIC    | input parameters for core part of the code (MHD, resistive wall, flow, etc..) |
| FEEDBACK | parameters for feedback control and linear plasma response                    |
| KINETIC  | parameters for including drift kinetic effects                                |
| QLIN     | parameters for quasi-linear initial value runs                                |
| NUMERIC  | numerical parameters of the code                                              |
| OUTOPT   | parameters controlling output options                                         |

## 4 MARS-\* input

The input to MARS-\* generally consists of three parts: (i) the CHEASE output files (Sec. 4.1); (ii) the Namelist input (Sec. 4.2); (iii) additional equilibrium radial profiles (Sec. 4.3).

### 4.1 Direct input data from CHEASE

As mentioned before, the direct input data from CHEASE into MARS-\* are two ascii files **OUTRMAR** and **OUTVMAR**. **OUTRMAR** contains all the necessary metrics information on the equilibrium inside the plasma, whilst **OUTVMAR** contains data specifying vacuum geometry outside the plasma.

### 4.2 Namelist input

There are 6 input Namelists defined in MARS-\*. These are briefly described by Table 5.

The input parameters defined in these Namelists are listed in Tables 6-18, together with the **default value** as well as other possible options where applicable. Note that these values also indicate the type of the associated parameter.

Table 6: Input parameters in Namelist BASIC: part 1.

| parameter | values                                       | description                                                                                                                                                                                                                                                   |
|-----------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALPHAR    | <b>2.</b>                                    | for analytic specification of fluid rotation profile ( <a href="#">NPROFR=2</a> )                                                                                                                                                                             |
| BETAR     | <b>1.</b>                                    | for analytic specification of fluid rotation profile ( <a href="#">NPROFR=2</a> )                                                                                                                                                                             |
| B0K       | <b>1.</b>                                    | (normalized) $B_0$ value used to define particle pitch angle $\Lambda$                                                                                                                                                                                        |
| CSROT0    | <b>1.</b>                                    | for specifying fluid rotation profile ( <a href="#">NPROFR=1,2</a> )                                                                                                                                                                                          |
| DEXQ      | <b>-2.</b>                                   | for specifying plasma density profile ( <a href="#">NPROFN=3</a> )                                                                                                                                                                                            |
| DINERT    | <b>1.</b>                                    | multiplier for LHS of equation for $\mathbf{v}$                                                                                                                                                                                                               |
| ETA       | <b>0.</b>                                    | 1/S at mag.axis, S=Lundquist number                                                                                                                                                                                                                           |
| ETAI      | <b>1.5</b>                                   | for specifying plasma density ( <a href="#">NPROFN=2</a> ) or resistivity ( <a href="#">NPROFT=2</a> ) profile                                                                                                                                                |
| ETASTE    | <b>1.</b>                                    | for analytic calculation of thermal electron diamagnetic frequency                                                                                                                                                                                            |
| ETASTI    | <b>1.</b>                                    | for analytic calculation of thermal ion diamagnetic frequency                                                                                                                                                                                                 |
| FRACPTH   | <b>1.</b>                                    | ratio of thermal to total equilibrium pressure at mag.axis                                                                                                                                                                                                    |
| GAMMA     | <b>5./3.</b>                                 | ratio of specific heats (amplitude)                                                                                                                                                                                                                           |
| GAMQ0     | <b>0.</b>                                    | Gaussian centre for radial profile of GAMMA, <a href="#">NPROFG=1</a>                                                                                                                                                                                         |
| GAMWID    | <b>1.E9</b>                                  | Gaussian width for radial profile of GAMMA, <a href="#">NPROFG=1</a>                                                                                                                                                                                          |
| IDIAMB    | <b>0</b><br>1-4<br>6                         | Ohm's law in standard single fluid MHD<br>include various terms associated with $P_e$ in extended Ohm's law<br>replace ion flow by electron flow in Ohm's law                                                                                                 |
| IDIAMTE   | <b>1</b><br><b>2</b>                         | instantaneous electron temperature equilibration<br>standard fluid equation for thermal electron pressure (similar to ions)                                                                                                                                   |
| IDIAMTI   | <b>0</b><br><b>1</b>                         | standard single fluid option for perturbed ion pressure<br>add Weiland's reactive closure term to perturbed thermal ion pressure equation                                                                                                                     |
| IDIAMV    | <b>0</b><br>1-3                              | no thermal ion diamagnetic flow included (standard MHD)<br>include various terms associated with thermal ion diamagnetic flow                                                                                                                                 |
| INERT1    | <b>.T.</b>                                   | include first part of Coriolis force due to toroidal flow                                                                                                                                                                                                     |
| INERT2    | <b>.T.</b>                                   | include second part of Coriolis force due to toroidal flow                                                                                                                                                                                                    |
| INERT3    | <b>.T.</b>                                   | include centrifugal force due to toroidal flow                                                                                                                                                                                                                |
| IVISC     | <b>0</b><br><b>1</b><br><b>2</b>             | no inclusion of parallel sound wave damping model<br>include full parallel sound wave damping model<br>displacement terms dropped from parallel sound wave damping model;<br>physically valid at vanishing flow, otherwise useful for numerical smoothing     |
| IWALL     | <b>0</b>                                     | array of cell numbers specifying wall position                                                                                                                                                                                                                |
| IWO       | <b>0</b><br><b>1</b><br><b>2</b><br><b>4</b> | old wall equation for complete thin wall approximation<br>based on field jump conditions<br>best wall equation for complete thin wall approximation<br>thin wall model with varying conductivity along poloidal angle<br>thin wall model with poloidal gap(s) |
| KGAM      | <b>10</b>                                    | for specifying local constant GAMMA ( <a href="#">NPROFG=2</a> ) or local constant fluid rotation ( <a href="#">NPROFR=3</a> ) near rational surfaces                                                                                                         |



Table 7: Input parameters in Namelist BASIC: part 2.

| parameter | values                               | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KRATSURF  | 1                                    | specifies $k$ -th rational surface for $\Delta'$ computations using the basis function method ( <b>INCFEED</b> =20)                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| MRATSURF  | 0<br>1<br>2                          | do not shift radial mesh points<br>shift integer radial mesh points to align exactly with rational surfaces<br>shift half-integer radial mesh points to align exactly with rational surfaces                                                                                                                                                                                                                                                                                                                                                                                          |
| M1        | 2                                    | lower bound for poloidal harmonic numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| M2        | 2                                    | upper bound for poloidal harmonic numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| NCASE     | 1<br>2<br>3<br>4<br>5<br>6<br>7<br>8 | run eigenvalue computation<br>run linear steady state response computation<br>linear response time-stepping with sinusoidal time varying current source<br>run three-stage feedback time-stepping simulation with possible voltage/current saturation<br>MARS-Q run with explicit time-stepping, <b>NSWEEP</b> =1, <b>NITMAX</b> =maximal time steps<br>MARS-Q run with implicit time-stepping, <b>NSWEEP</b> =maximal time steps, <b>NITMAX</b> =1<br>assume spontaneous plasma response at each time step in MARS-Q runs<br>A benchmark test case for the momentum solver in MARS-Q |
| NCOUPPL   | 0<br>-3,-1,-2,1-9                    | no coupling to CARIDDI code<br>various coupling schemes to CARIDDI to form CarMa code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| NEXPV     | 0<br>1                               | read in only radial profiles (amplitude normalized to 1 at magnetic axis) from experimental data for plasma density, rotation, temperature etc.<br>read in both amplitudes and radial profiles from experimental data files for plasma density, rotation, temperature etc..                                                                                                                                                                                                                                                                                                           |
| NITMAX    | 50                                   | maximal iteration for eigenvalue solver or explicit initial value solver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| NPROFG    | 0<br>1<br>2                          | constant ratio of specific heats<br>analytic Gaussian model for the GAMMA profile<br>piece-wise constant GAMMA (near rational surfaces)                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| NPROFIE   | 0<br>4                               | thermal ion and electron temperature profiles computed as ratio of pressure to density<br>read in experimental temperature profiles <b>PROFTL.IN</b> and <b>PROFTE.IN</b>                                                                                                                                                                                                                                                                                                                                                                                                             |
| NPROFN    | 0<br>1<br>2<br>3<br>4                | analytic specification of plasma density profile in volume-based radial grid<br>analytic specification of plasma density profile in $\psi$ -based radial grid<br>analytic specification of plasma density profile via equilibrium pressure profile<br>analytic specification of plasma density profile via $q$ -profile<br>read in experimental plasma density profile <b>PROFDEN.IN</b>                                                                                                                                                                                              |
| NPROFR    | 0<br>1<br>2<br>3<br>4                | uniform fluid rotation profile<br>analytic specification of fluid rotation profile in $\psi$ -based radial grid<br>analytic specification of fluid rotation profile via equilibrium pressure profile<br>analytic specification of fluid rotation profile near rational surfaces<br>read in experimental fluid rotation profile <b>PROFROT.IN</b>                                                                                                                                                                                                                                      |



Table 8: Input parameters in Namelist BASIC: part 3.

| parameter                 | values                       | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">NPROFT</a>    | <b>0</b><br>1<br>2<br>3<br>5 | analytic specification of plasma resistivity profile in volume-based radial grid<br>analytic specification of plasma resistivity profile in $\psi$ -based radial grid<br>analytic specification of plasma resistivity profile via equilibrium pressure profile<br>analytic specification of plasma resistivity profile via $T_e$ -profile: Spitzer model<br>analytic specification of plasma resistivity profile for $\Delta'$ calculations with multiple rational surfaces |
| <a href="#">NPROFTTCA</a> | <b>0</b><br>4                | constant parallel thermal transport coefficient<br>read in experimental profile for parallel thermal transport coefficient<br><a href="#">PROFTTCPARA.IN</a>                                                                                                                                                                                                                                                                                                                |
| <a href="#">NPROFTTCE</a> | <b>0</b><br>4                | constant perpendicular thermal transport coefficient<br>read in experimental profile for perpendicular thermal transport coefficient<br><a href="#">PROFTTCEPERP.IN</a>                                                                                                                                                                                                                                                                                                     |
| <a href="#">NPROFWE</a>   | <b>0</b><br>4                | uniform $E \times B$ rotation profile<br>read in experimental $E \times B$ rotation profile <a href="#">PROFWE.IN</a>                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">NSWEEP</a>    | <b>1</b>                     | number of outer loop iterations for NCASE=4,6,7, or feedback gain scan                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">NV</a>        | <b>0</b>                     | cell number specifying vacuum extension                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <a href="#">NWALL</a>     | <b>0</b>                     | number of resistive walls                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">OMEGA0</a>    | <b>0.</b>                    | for specifying fluid rotation profile ( <a href="#">NPROFR</a> =1,2)                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">OMEGA1</a>    | <b>0.</b>                    | for specifying fluid rotation profile ( <a href="#">NPROFR</a> =1,2)                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">PVISC</a>     | <b>0.</b>                    | parallel sound wave damping coefficient                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <a href="#">RHO1</a>      | <b>0.</b>                    | for specifying plasma density profile ( <a href="#">NPROFN</a> =0,1)                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">RHO2</a>      | <b>0.</b>                    | for specifying plasma density profile ( <a href="#">NPROFN</a> =0,1)                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">RNTOR</a>     | <b>-1.</b>                   | toroidal mode number                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">ROTE</a>      | <b>1.E-2</b>                 | amplitude of equilibrium fluid rotation frequency                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <a href="#">ROTWE0</a>    | <b>0.</b>                    | amplitude of equilibrium EXB rotation frequency                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <a href="#">SIGMA1</a>    | <b>0.</b>                    | for specifying plasma resistivity profile ( <a href="#">NPROFT</a> =0,1,5)                                                                                                                                                                                                                                                                                                                                                                                                  |
| <a href="#">SIGMA2</a>    | <b>0.</b>                    | for specifying plasma resistivity profile ( <a href="#">NPROFT</a> =0,1,5)                                                                                                                                                                                                                                                                                                                                                                                                  |
| <a href="#">TALPHA1</a>   | <b>(1.,0.)</b>               | initial guess for eigenvalue problems;<br>for MARS-Q: 1/RE=initial time step, IM=excitation frequency                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">TAUW</a>      | <b>2.E4</b>                  | assumed wall time (amplitude), array                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">TTCCONV0</a>  | <b>1.</b>                    | multiplier for convective term of equation of state                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <a href="#">TTCPARA0</a>  | <b>0.</b>                    | amplitude of parallel thermal conduction coefficient in equation of state                                                                                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">TTCPERP0</a>  | <b>0.</b>                    | amplitude of perpendicular thermal conduction coefficient in equation of state                                                                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">ZCHARGE</a>   | <b>1</b>                     | particle charge number in MARS-Q NTV module                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Table 9: Input parameters in Namelist FEEDBACK: part 1.

| parameter | values                                                                                                                                                                                 | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BTCHI     | <b>0.06</b>                                                                                                                                                                            | poloidal angle specifying thickness of active coil's leg, normalized by $\pi$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| DTRAMPUI  | <b>-1.</b>                                                                                                                                                                             | linear coil current ramp-up time for response computations<br>negative value means instantaneous ramp-up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| EXTCURR   | <b>0.</b>                                                                                                                                                                              | specify starting and ending time steps as well as excitation frequency of coil current, for linear initial value response computations ( <b>NCASE=3</b> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| FCCHI     | <b>0.</b>                                                                                                                                                                              | poloidal angle specifying centre of active coil, normalized by $\pi$ , array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| FEEDI     | <b>(0.,0.)</b>                                                                                                                                                                         | base coil current for feedback and response computations, array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| FEEDIT    | <b>(0.,0.)</b>                                                                                                                                                                         | toroidal array of coil currents, of size array <b>NCOILT</b> , for CarMa runs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| FNUD      | <b>(0.,0.)</b>                                                                                                                                                                         | coefficients for derivative gain, array of size <b>NSENS+1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| FNUP      | <b>(0.,0.)</b>                                                                                                                                                                         | coefficients for proportional gain, array of size <b>NSENS+1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| FWCHI     | <b>0.</b>                                                                                                                                                                              | poloidal angle specifying width of active coil, normalized by $\pi$ , array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| GAINA     | <b>0.</b>                                                                                                                                                                              | feedback gain amplitude, array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| GAINP     | <b>0.</b>                                                                                                                                                                              | feedback gain phase, array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| IBOUT     | <b>2</b>                                                                                                                                                                               | specify radial cell number where to output field <b>B123U.OUT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| IBOUT2    | <b>3</b>                                                                                                                                                                               | specify radial cell number where to output field <b>B123U2.OUT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| IDYNAM    | <b>0</b><br><b>1</b><br><b>2</b>                                                                                                                                                       | current control<br>voltage control with resistive-inductive active coils<br>voltage control with super-conducting active coils                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| INCFEED   | <b>0</b><br><b>1</b><br><b>2</b><br><b>3</b><br><b>4</b><br><b>5</b><br><b>6</b><br><b>8</b><br><b>9</b><br><b>10</b><br><b>11</b><br><b>12</b><br><b>20</b><br><b>21</b><br><b>22</b> | no feedback included<br>feedback with radial flux sensor<br>compute wall decay time in free space ( <b>FEEDI=0</b> )<br>feedback with point-wise poloidal field sensor<br>compute vacuum field due to external current sources (w/ or w/o wall response)<br>feedback with point-wise toroidal field sensor<br>magnetic field perturbation due to a given wall current alone<br>compute plasma response to external current sources (w/ or w/o wall response)<br>“vacuum” feedback with point-wise poloidal sensor<br>vacuum response to external current sources<br>assuming plasma as rigid ideal conductor<br>feedback with point-wise radial field sensor<br>“vacuum” feedback with point-wise radial sensor<br>computes $\Delta'$ using basis function approach: small solution response<br>computes $\Delta'$ using basis function approach: large solution response<br>vacuum response to a fixed radial field at computational boundary |
| ISENS     | <b>IFEED,0</b>                                                                                                                                                                         | array of cell numbers specifying radial location of sensor coils<br>array of form <b>[0:NSENS]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Table 10: Input parameters in Namelist FEEDBACK: part 2.

| parameter        | values                    | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>KKF</b>       | -3<br>-2<br>-1<br>0<br>>0 | read in radial field perturbation from input file <b>INPUT_BNM.IN</b> ,<br>compute vacuum equivalent surface current at surface <b>IFEED</b><br>read in external surface current source (for poloidal component) at surface <b>IFEED</b> , from input file <b>CURHARMO.IN</b><br>Gaussian-representation for toroidal coil current,<br>with analytic calculation of poloidal coil current<br>Heaviside-representation for poloidal coil current<br>specify power factor of generalized Gaussian-representation for toroidal<br>coil current, with numerical computation of poloidal coil current |
| <b>KREADECA</b>  | 0<br>1                    | compute response matrix for equivalent surface current<br>representation of error fields, based on Biot-Savart law<br>read in response matrix <b>ECA.IN</b> for equivalent surface current approach                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>NCOIL</b>     | 1                         | number of active and sensor coils along poloidal angle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>NCOILT</b>    | 6                         | number of active coils along toroidal angle, for CarMa runs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>NSENS</b>     | 0                         | number of sensor coil arrays along minor radius, counting from 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>NSENST</b>    | 6                         | number of sensor coils along toroidal angle, for CarMa runs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>PHIPHASE</b>  | 0.                        | specify phase factor of coil currents for CarMa runs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>PHISIGN</b>   | -1.                       | specify direction of $\phi$ -angle for CarMa runs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>RL0</b>       | 1.                        | self-inductance of active coil, for voltage control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>RMFS</b>      | 1.                        | free-space mutual inductance between active coils and sensors,<br>to be used to normalize feedback gain, array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>SCCHI</b>     | 0.                        | poloidal angle specifying centre of sensor coil,<br>normalized by $\pi$ , array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>SWCHI</b>     | 0.                        | poloidal angle specifying width of radial flux sensor coil,<br>normalized by $\pi$ , array of size <b>NCOIL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>TAUF</b>      | 2.E+4                     | L/R time of feedback coils, for voltage control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>THRESHOLD</b> | 1.                        | threshold field amplitude for non-linear feedback<br>( <b>NCASE</b> =4) or certain MARS-Q runs ( <b>NCASE</b> =6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Table 11: Input parameters in Namelist KINETIC: part 1.

| parameter  | values            | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALPHAD     | 1.                | global multiplier for all kinetic contributions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ALPHAH     | 1.                | multiplier for all energetic particle pressure fraction data read from <b>PROFPA.IN</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ALPHAP     | 0.5               | ratio of thermal ion temperature to total thermal temperature at magnetic axis                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ALTAU      | (0.5,0.)          | relaxation parameter for outer loop iteration of non-linear eigenvalue solver                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ATAU       | (1.E+11,0.)       | mode frequency entering into kinetic resonance operators, replaced by self-consistently computed mode frequency if $ ATAU  > 1.E+10$                                                                                                                                                                                                                                                                                                                                                                                              |
| DZETA0     | $\sqrt{0.015}$    | width of Gaussian at birth energy in NBI model <b>ISPECIES_F0</b> =3 for energetic particles with anisotropic pitch angle distribution                                                                                                                                                                                                                                                                                                                                                                                            |
| EP_DENF0   | 0.05              | on-axis density fraction for energetic particles: <b>NPROFK</b> =0,1                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| EP_PREF0   | 0.5               | on-axis pressure fraction for energetic particles: <b>NPROFK</b> =0,1                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ESPECIES_M | 2.                | mass number of all particle species (default 5.4463E-4 for electrons)                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ESPECIES_Z | 1.,-1.,1.         | charge number of all particle species                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| HHTYPE4C   | 0.5               | normalized equilibrium field strength corresponding to ICRH resonance, for energetic particles with ICRH model <b>ISPECIES_F0</b> =4                                                                                                                                                                                                                                                                                                                                                                                              |
| IFOW       | 0<br>1            | drift orbit tracing for a test particle, with vanishing orbit width<br>full drift orbit tracing for a test particle                                                                                                                                                                                                                                                                                                                                                                                                               |
| IFOWPSI0   | 1<br>2            | $\hat{\psi}$ -approach for including FOW corrections, that corresponds to neoclassical theory for bootstrap current<br>$\tilde{\psi}$ -approach for including FOW corrections, that does not correspond to neoclassical theory                                                                                                                                                                                                                                                                                                    |
| IGAMMADIVV | 0<br>1            | keep equilibrium pressure in compressibility term of perturbed equation of state<br>keep sum of pressures from particle species w/o non-adiabatic kinetic contributions in the compressibility term                                                                                                                                                                                                                                                                                                                               |
| INCKIN     | 0<br>1<br>2       | no drift kinetic computations included<br>includes drift kinetic computations<br>fluid-wise implementation of kinetic contributions at CGL limit                                                                                                                                                                                                                                                                                                                                                                                  |
| INUTYPE    | -1<br>0<br>1<br>2 | numerical integration over thermal particle energy space with energy-independent Krook collisionality coefficients<br>analytic integration over thermal particle energy space with energy-independent Krook collisionality coefficients<br>numerical integration over thermal particle energy space with Krook collisionality coefficients depending on both particle energy and harmonic numbers<br>numerical integration over thermal particle energy space with Krook collisionality coefficients depending on particle energy |
| IORBIT     | 0<br>1            | do not perform drift orbit tracing for a test particle<br>perform drift orbit tracing for a test particle                                                                                                                                                                                                                                                                                                                                                                                                                         |

Table 12: Input parameters in Namelist KINETIC: part 2.

| parameter   | values | description                                                                                                                 |
|-------------|--------|-----------------------------------------------------------------------------------------------------------------------------|
| IPERTURB    | 0      | non-perturbative MHD-kinetic hybrid approach                                                                                |
|             | 1      | perturbative MHD-kinetic hybrid approach with flow included into fluid part                                                 |
|             | 2      | perturbative MHD-kinetic hybrid approach with flow excluded from fluid part                                                 |
| ISPECIES_F0 | 0      | array specifying types of particle species as defined in the code                                                           |
| KEFORM      | 0      | do not compute and output perturbed energy components                                                                       |
|             | 1      | compute energy components from non-quadratic forms                                                                          |
|             | 2      | compute energy components using quadratic forms                                                                             |
| KENORM      | 1      | normalize all energy components by full plasma inertia energy                                                               |
|             | 2      | normalize all energy components by plasma inertia energy associated with radial displacement only                           |
| KFASTRUN    | 0      | numerical integration over particle energy space, for all particles                                                         |
|             | 1      | analytic integration, where possible, over particle energy space                                                            |
| KNBI        | 0      | normal injection NBI model: ZZETA0=0.                                                                                       |
|             | 2      | symmetric co- and counter injections NBI model                                                                              |
|             | 11     | co-tangential injection NBI model by Gorelenkov                                                                             |
|             | 12     | co-tangential injection NBI model maintaining continuity in pitch angle space                                               |
|             | 13     | co-tangential injection NBI model maintaining continuity in pitch angle space, and always positive-valued                   |
|             | 31     | counter-tangential injection NBI model by Gorelenkov                                                                        |
|             | 32     | counter-tangential injection NBI model maintaining continuity in pitch angle space                                          |
|             | 33     | counter-tangential injection NBI model maintaining continuity in pitch angle space, and always positive-valued              |
| KSMTYPE4    | 1      | parabolic model for pitch angle dependent asymmetry factor in ICRH model for energetic particles ISPECIES_F0=4              |
|             | 2      | exponential model for pitch angle dependent asymmetry factor in ICRH model for energetic particles ISPECIES_F0=4            |
| NNSCAN      | 0      | save intermediate data for distribution function ISPECIES_F0=3                                                              |
|             | 1      | read intermediate data for distribution function ISPECIES_F0=3                                                              |
| NORB        | 1000   | number of time steps for drift orbit tracing of a test particle                                                             |
| NPROFK      | 0      | constant equilibrium density and pressure fractions for energetic particles                                                 |
|             | 1      | analytic specification of radial profiles for equilibrium density and pressure fractions for energetic particles            |
|             | 4      | read in experimental radial profile for energetic density fraction from PROFDA.IN and pressure fraction from PROFPA.IN      |
| NPROFR4     | 0      | constant ratio of perpendicular to parallel equilibrium temperatures, for energetic particles with ICRH model ISPECIES_F0=4 |
|             | 4      | read in experimental radial profile from PROFR4.IN                                                                          |

Table 13: Input parameters in Namelist KINETIC: part 3.

| parameter     | values | description                                                                                                                                                                                                  |
|---------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NPROFS4       | 0<br>4 | constant asymmetry factor of equilibrium distribution function for passing energetic particles with ICRH model <code>ISPECIES_F0=4</code><br>read in experimental radial profile from <code>PROFS4.IN</code> |
| NPROFUE       | 0<br>4 | constant effective thermal electron Krook collision frequency<br>read in experimental radial profile from <code>PROFNUE.IN</code>                                                                            |
| NPROFUI       | 0<br>4 | constant effective thermal ion Krook collision frequency<br>read in experimental radial profile from <code>PROFNUI.IN</code>                                                                                 |
| NSPECIES      | 2      | number of particle species participating in kinetic computations                                                                                                                                             |
| NUEFFEA       | 0.     | effective thermal electron Krook collision frequency at magnetic axis                                                                                                                                        |
| NUEFFIA       | 0.     | effective thermal ion Krook collision frequency at magnetic axis                                                                                                                                             |
| NUMODEL       | 0<br>1 | default Krook collisionality model<br>Krook collisionality with another normalization                                                                                                                        |
| OMEGACI0      | 50.    | thermal ion gyro-frequency normalized by Alfvén frequency, at magnetic axis                                                                                                                                  |
| ORBE          | 100.   | for computing full drift orbit of a test particle:<br>normalized particle energy                                                                                                                             |
| ORBL          | 0.25   | for computing full drift orbit of a test particle:<br>particle pitch angle                                                                                                                                   |
| ORBSIGINI     | 1.     | for computing full drift orbit of a test particle:<br>initial sign of parallel speed                                                                                                                         |
| PSPECIES_AP   | 0.     | array multiplier specifying adiabatic contribution<br>from passing particles, for all particle species                                                                                                       |
| PSPECIES_AT   | 0.     | array multiplier specifying adiabatic contribution<br>from trapped particles, for all particle species                                                                                                       |
| PSPECIES_FOWP | 0.     | array multiplier specifying kinetic contribution due to<br>FOW correction from passing particles, for all particle species                                                                                   |
| PSPECIES_FOWT | 0.     | array multiplier specifying kinetic contribution due to<br>FOW correction from trapped particles, for all particle species                                                                                   |
| PSPECIES_NP   | 0.     | array multiplier specifying non-adiabatic contribution<br>from transit resonance of passing particles, for all particle species                                                                              |
| PSPECIES_NTB  | 0.     | array multiplier specifying non-adiabatic contribution<br>from bounce resonance of trapped particles, for all particle species                                                                               |
| PSPECIES_NTD  | 0.     | array multiplier specifying non-adiabatic contribution<br>from precession resonance of trapped particles, for all particle species                                                                           |
| RTAN          | 0.5    | NBI injection tangency radii                                                                                                                                                                                 |
| R0TYPE4       | 1.     | ratio of perpendicular to parallel equilibrium temperatures<br>at magnetic axis, for energetic particles with<br>ICRH model <code>ISPECIES_F0=4</code>                                                       |
| S0TYPE4       | 0.     | asymmetry factor, at magnetic axis, of equilibrium distribution<br>function for passing energetic particles with<br>ICRH model <code>ISPECIES_F0=4</code>                                                    |
| ZZETA0        | 0.8    | centre of Gaussian in NBI model <code>ISPECIES_F0=3</code><br>for energetic particles with anisotropic pitch angle distribution                                                                              |

Table 14: Input parameters in Namelist QLIN: part 1.

| parameter | values   | description                                                                                        |
|-----------|----------|----------------------------------------------------------------------------------------------------|
| CTEDGE    | 1.       | minor radius up to which densities of various torque and energy components are computed            |
| CTJXB     | 0.       | multiplier for jxb torque to be included in $n = 0$ momentum equation                              |
| CTNTV     | 0.       | multiplier for NTV torque to be included in $n = 0$ momentum equation                              |
| CTREY     | 0.       | multiplier for Reynolds torque to be included in $n = 0$ momentum equation                         |
| ITSATURAT | 0        | Neumann BC for $n = 0$ momentum equation at plasma boundary                                        |
|           | 1        | total rotation not allowed to cross zero                                                           |
|           | 2        | Dirichlet BC for $n = 0$ momentum equation at plasma boundary                                      |
| KSOLREAD  | 0        | read in full solution from <b>MARSQ.SOL</b> as initial condition                                   |
| KSOLSAVE  | 0        | write out full solution to <b>MARSQ.SOL</b>                                                        |
| NPROFVD   | 0        | constant momentum diffusion coefficient                                                            |
|           | 1        | analytic specification of radial profile for momentum diffusion coefficient: power of minor radius |
|           | 2        | analytic specification of radial profile for momentum diffusion coefficient: $T_e^{-3/2}/s$        |
|           | 3        | analytic specification of radial profile for momentum diffusion coefficient: $T_e^{-3/2}$          |
|           | 4        | read in experimental profile for momentum diffusion from <b>PROFTVD.IN</b>                         |
| NPROFVP   | 0        | constant coefficient for momentum pinch                                                            |
|           | 4        | read in experimental profile for momentum pinch from <b>PROFTVP.IN</b>                             |
| TALPHA2   | 1.       | implicit coefficient for MHD solver with equilibrium flow, $= \alpha_2$ in Table 15                |
| TALPHA3   | 1.       | implicit coefficient for MHD terms with flow correction, $= \alpha_3$ in Table 15                  |
| TALPHA4   | (-1.,0.) | algorithm switch between eigen/initial value solvers, $= \alpha_4$ in Table 15                     |
| TALPHA5   | (1.,0.)  | set time step for $n = 0$ momentum solver, $= \alpha_5$ in Table 15                                |
| TALPHA6   | 1.       | implicit coefficient for linear terms in momentum solver, $= \alpha_6$ in Table 15                 |
| TALPHA7   | 0.       | multiplier for quadratic terms in momentum solver, $= \alpha_7$ in Table 15                        |
| TALPHA8   | 0.9      | correcting multiplier for time step in adaptive scheme                                             |
| TCHIM0    | 1.E-7    | amplitude of momentum diffusion coefficient                                                        |
| TDELTALIM | 100.     | overall upper bound for time step                                                                  |
| TDELTALOW | 1.1      | overall lower bound for time step                                                                  |
| TDELTAMAX | 0.08     | upper limit in solution change for triggering time step correction (decrease)                      |
| TDELTAMIN | 0.04     | lower limit in solution change for triggering time step correction (increase)                      |
| TPOWVD    | 0.       | power factor in analytic specification of momentum diffusion coefficient with <b>NPROFVD</b> =1    |
| TVPINCH0  | 0.       | amplitude of momentum pinch coefficient                                                            |



Table 15: Input parameters in Namelist QLIN: part 2.

| NCASE | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$    | $\alpha_5$ | $\alpha_6$ | $\alpha_7$ | NSWEEP   | NITMAX |
|-------|------------|------------|------------|---------------|------------|------------|------------|----------|--------|
| 1     | $\gamma_0$ | 1          | 1          | $-1/\gamma_0$ |            |            | 0          | $\geq 1$ | $> 1$  |
| 2     | $\gamma_0$ | 1          | 1          | 0             |            |            | 0          | 1        | 1      |
| 3     | $1/\tau_0$ | (0.5,1]    |            | 1             |            |            | 0          | 1        | $> 1$  |
| 4     | $\gamma_0$ | 1          | 1          | $-1/\gamma_0$ |            |            | 0          | 3        | $> 1$  |
|       | $2\gamma$  | 1          |            | 1             |            |            | 0          | 3        | $> 1$  |
|       | $2\gamma$  | 1          |            | 1             |            |            | 0          | 3        | $> 1$  |
| 5     | $1/\tau_0$ | (0.5,1]    | 0          | 1             | $\alpha_1$ | (0.5,1]    | 1          | 1        | $> 1$  |
| 6     | $1/\tau_0$ | (0.5,1]    | (0.5,1]    | 1             | $\alpha_1$ | (0.5,1]    | 1          | $> 1$    | 1      |
| 7     | $\gamma_0$ | 1          | 1          | 0             | $1/\tau_0$ | (0.5,1]    | 1          | $> 1$    | 1      |
| 8     |            |            |            |               | $1/\tau_0$ | (0.5,1]    | 1          | 1        | $> 1$  |

#### 4.2.1 Namelist BASIC

#### 4.2.2 Namelist FEEDBACK

#### 4.2.3 Namelist KINETIC

#### 4.2.4 Namelist QLIN

#### 4.2.5 Namelist NUMERIC

#### 4.2.6 Namelist OUTOPT

### 4.3 Specification of additional equilibrium radial profiles

The CHEASE run does not give all equilibrium radial profiles required by MARS-\*. Specification of key additional profiles are described below, either analytically or numerically. Most of the 1D equilibrium profiles are stored in the MARS-\* output file [PROFEQ.OUT](#).

#### 4.3.1 Analytic specification

Shown below are those equilibrium quantities for which multiple choices of analytic specification of the radial profiles are available. The radial variables  $\psi_v, \psi_p, \psi_t$  denote the plasma volume, equilibrium poloidal magnetic flux, equilibrium toroidal magnetic flux, respectively, enclosed by a given flux surface. These quantities are all equal to 0 at the magnetic axis, and normalized to 1 at the plasma boundary.



Table 16: Input parameters in Namelist NUMERIC: part 1.

| parameter | values                           | description                                                                                                                                                                                                                             |
|-----------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADAPSDF   | <b>0.2</b>                       | weighting factor in radial mesh adaptivity:<br>for segment length associated with objective function                                                                                                                                    |
| ADAPSD2   | <b>0.</b>                        | weighting factor in radial mesh adaptivity:<br>for inverse curvature associated with objective function                                                                                                                                 |
| ADAPSLF   | <b>0.2</b>                       | overall weighting factor in radial mesh adaptivity                                                                                                                                                                                      |
| ADAPSNU   | <b>0.5</b>                       | smoothing factor in generating control function for radial mesh adaptivity                                                                                                                                                              |
| DELRATS   | <b>0.</b>                        | width of window functions for annihilating kinetic contributions<br>near rational surfaces (no annihilation by default)                                                                                                                 |
| EPSDET    | <b>1.E-199</b>                   | tolerance for singularity of determinant of system matrices                                                                                                                                                                             |
| EPSPAM    | <b>1.E-7</b>                     | tolerance for iteration convergence of eigenvalue solver                                                                                                                                                                                |
| EQFAC     | <b>1.</b>                        | multiplier to both LHS and RHS of subset of equations                                                                                                                                                                                   |
| IOMPNUM   | <b>1</b>                         | number of processors for OpenMP                                                                                                                                                                                                         |
| IPDIVB    | <b>0</b><br><b>1</b><br><b>2</b> | divergence-free condition for magnetic field not explicitly enforced<br>divergence-free condition for magnetic field explicitly enforced for $n \neq 0$<br>divergence-free condition for magnetic field explicitly enforced for $n = 0$ |
| KADAPS    | <b>0</b><br><b>1</b>             | normal run<br>read in new radial mesh from <b>PROFEQ.OUT</b>                                                                                                                                                                            |
| NADAPS    | <b>0</b>                         | maximal number of adaptive radial mesh refinements                                                                                                                                                                                      |
| NCHIO     | <b>120</b>                       | mesh size of poloidal angle for computing particle phase factors                                                                                                                                                                        |
| NCONVB1   | <b>0</b><br><b>1</b><br><b>2</b> | normal runs<br>compute and save poloidal angle vs. geometric angle at rational surfaces<br>read in poloidal angle data for transferring radial field from one coordinate<br>system to PEST-like co-ordinates, at rational surfaces      |
| NCONVCS   | <b>0</b><br><b>1</b>             | normal runs<br>transfer CHEASE generated coordinate system into geometric poloidal<br>angle based coordinate system before running MARS-*                                                                                               |
| NEPK      | <b>101</b>                       | mesh size for particle kinetic energy                                                                                                                                                                                                   |
| NFIT      | <b>3</b>                         | Neumann condition for $ m  = 1$ components of radial displacement/velocity<br>at several points near magnetic axis                                                                                                                      |
| NKL0      | <b>0</b>                         | extension of bounce harmonic coverage over poloidal harmonics                                                                                                                                                                           |
| NKSMOOTHB | <b>20</b>                        | number of poloidal harmonics for smoothing equilibrium mod-B<br>in drift kinetic computations                                                                                                                                           |
| NKSMOOTH  | <b>10</b>                        | iteration number of diffusion-based smoothing for radial profiles<br>of equilibrium quantities used in drift kinetic computations                                                                                                       |
| NLAMIN    | <b>100</b>                       | lower bound of mesh size for trapped particle pitch angle                                                                                                                                                                               |
| NLAMK     | <b>201</b>                       | (maximal) mesh size for particle pitch angle                                                                                                                                                                                            |
| NRES      | <b>6</b>                         | patch resistivity near magnetic axis, works only if <b>NFIT</b> < <b>NRES</b>                                                                                                                                                           |
| NTORQ     | <b>1</b>                         | enforce constant torque densities near magnetic axis                                                                                                                                                                                    |
| NVACJ     | <b>0</b><br><b>1</b>             | normal runs<br>multiply vacuum, coil and wall equations by Jacobian for $n \neq 0$ ; can be<br>beneficial for response computations where vacuum domain<br>extends into $R < 0$ region                                                  |

Table 17: Input parameters in Namelist NUMERIC: part 2.

| parameter              | values                  | description                                                                                                                                                                                                                                                                  |
|------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">PTRAPH</a> | <b>0.9</b>              | weighting factor in tunable integration for half-integer grid based variables                                                                                                                                                                                                |
| <a href="#">PTRAPI</a> | <b>1.</b>               | weighting factor in tunable integration for integer grid based variables                                                                                                                                                                                                     |
| <a href="#">V2XKEY</a> | <b>0</b><br>1<br>2<br>3 | perturbed velocity is used in both fluid and kinetic pressures<br>perturbed velocity (displacement) is used in fluid (kinetic) pressure<br>perturbed velocity (displacement) is used in kinetic (fluid) pressure<br>displacement is used in both fluid and kinetic pressures |

Table 18: Input parameters in Namelist OUTOPT

| parameter               | values      | description                                                                              |
|-------------------------|-------------|------------------------------------------------------------------------------------------|
| <a href="#">DCONTI</a>  | <b>.F.</b>  | locate continuum singularity                                                             |
| <a href="#">DPLOLP</a>  | <b>.F.</b>  | array controlling on-screen printing of eigenfunctions                                   |
| <a href="#">DPRINT</a>  | <b>.F.</b>  | print eigenvectors on screen for debugging                                               |
| <a href="#">DTERMS</a>  | <b>.F.</b>  | diagnostics for resistive ballooning modes                                               |
| <a href="#">JSOUT</a>   | <b>10</b>   | radial cell number for outputting diagnostic quantities                                  |
| <a href="#">MOUTPUT</a> | <b>60</b>   | Number of poloidal harmonics for outputting equilibrium quantities                       |
| <a href="#">NORR</a>    | <b>0</b>    | R-mesh size of rectangular domain for outputting B-field and A-field                     |
| <a href="#">NOZZ</a>    | <b>0</b>    | Z-mesh size of rectangular domain for outputting B-field and A-field                     |
| <a href="#">ODWKCOM</a> | <b>.F.</b>  | compute and output all components of $\delta W_K$                                        |
| <a href="#">ORMIN</a>   | <b>0.6</b>  | specify rectangular R-Z domain for outputting B-field and A-field                        |
| <a href="#">ORMAX</a>   | <b>1.4</b>  | specify rectangular R-Z domain for outputting B-field and A-field                        |
| <a href="#">OUTDAT</a>  | <b>.F.</b>  | output basic solutions for further processing                                            |
| <a href="#">OZMIN</a>   | <b>-0.4</b> | specify rectangular R-Z domain for outputting B-field and A-field                        |
| <a href="#">OZMAX</a>   | <b>0.4</b>  | specify rectangular R-Z domain for outputting B-field and A-field                        |
| <a href="#">PLSURF</a>  | <b>.F.</b>  | compute orthogonal components of perturbed velocity and magnetic field at plasma surface |

Denoting  $\rho$ ="normalized plasma density",  $\rho_1$ =[RHO1](#),  $\rho_2$ =[RHO2](#), we have

$$\rho = \begin{cases} 1 + \psi_v \rho_1 + \psi_v^2 \rho_2 & (\text{NPROFN} = 0) \\ 1 + \psi_p \rho_1 + \psi_p^2 \rho_2 & (\text{NPROFN} = 1) \\ [P_{eq}/P_{eq}(0)]^{1/(1+\text{ETA1})} & (\text{NPROFN} = 2) \\ [q/q(0)]^{\text{DEXQ}} & (\text{NPROFN} = 3) \end{cases} \quad (20)$$

Denoting  $\Omega$ ="toroidal rotation frequency",  $\Omega_R$ =[ROTE](#),  $\Omega_0$ =[OMEGA0](#),  $\Omega_1$ =[OMEGA1](#),  $\psi_1$ =[CSROT0](#)<sup>2</sup>,  $\alpha_R$ =[ALPHAR](#),  $\beta_R$ =[BETAR](#), we have

$$\Omega = \begin{cases} \Omega_R & (\text{NPROFR} = 0) \\ \Omega_R [\Omega_1 + (\Omega_0 - \Omega_1) (1 - 2\psi_p/\psi_1 + \psi_p^2/\psi_1^2)] & (\text{NPROFR} = 1, \psi_p < \psi_1) \\ 0 & (\text{NPROFR} = 1, \psi_p > \psi_1) \\ \Omega_R [\Omega_1 + (\Omega_0 - \Omega_1) (1 - \psi_v^{\alpha_R/2})^{\beta_R}] & (\text{NPROFR} = 2) \\ \Omega_R & (\text{NPROFR} = 3, \text{within KGAM cells}) \\ 0 & (\text{NPROFR} = 3, \text{outside KGAM cells}) \end{cases} \quad (21)$$

Denoting  $\eta$ ="plasma resistivity",  $\eta_0$ =[ETA](#),  $\eta_1 = \eta_0/(R_0/a)^2$ ,  $\sigma_1$ =[SIGMA1](#),  $\sigma_2$ =[SIGMA2](#),  $\eta_i$ =[ETAI](#),  $\eta_p = -1.5\eta_i/(1 + \eta_i)$ , we have

$$\eta = \begin{cases} \eta_1/[1 + (\sigma_1 + \sigma_2\psi_v)\psi_v] & (\text{NPROFT} = 0) \\ \eta_1/[1 + (\sigma_1 + \sigma_2\psi_p)\psi_p] & (\text{NPROFT} = 1) \\ \eta_1[P_{eq}/P_{eq}(0)]^{\eta_p} & (\text{NPROFT} = 2) \\ \eta_1[T_e/T_e(0)]^{-3/2} & (\text{NPROFT} = 3) \end{cases} \quad (22)$$

Denoting  $\chi_M$ ="toroidal momentum diffusion coefficient",  $\chi_{M0}$ =[TCHIM0](#),  $\chi_p$ =[TPOWVD](#), we have

$$\chi_M = \begin{cases} \chi_{M0} & (\text{NPROFVD} = 0) \\ \chi_{M0}\psi_p^{\chi_p/2} & (\text{NPROFVD} = 1) \\ \chi_{M0}[T_e/T_e(0)]^{-3/2}/\psi_p^{1/2} & (\text{NPROFVD} = 2) \\ \chi_{M0}[T_e/T_e(0)]^{-3/2} & (\text{NPROFVD} = 3) \end{cases} \quad (23)$$

### 4.3.2 Specification via input data files

A uniform format for input profile data is adopted in MARS-\*

```
=====
N KEY
[x] [V]
=====
```

where N = the total number of data points = dimension of 1D arrays [x] and [V], KEY specifies the choice of the radial variable [x]: if KEY=1,  $[x]=s \equiv \sqrt{\psi_p}$ , where  $\psi_p$  is the normalised poloidal magnetic flux; if KEY=2,  $[x]=\sqrt{\psi_t}$ , with  $\psi_t$  being the normalised toroidal flux.

For all the input data but the toroidal rotation, MARS-\* reads in the data vector  $[V]$  and normalises it to unity at the magnetic axis. The amplitude of the data  $[V]$  is then controlled by the corresponding input Namelist parameter. For the toroidal rotation, the vector  $[V]$  is normalised such that  $\max[V]=1$ .

As shown in the MARS-\* input Namelist Tables, the code reads in (and thus requires) the input data file **PROF\***, if the corresponding Namelist control parameter **NPROF\*** is ascribed the option “4”. All the other options of **NPROF\*** determines analytical specification of the corresponding radial profiles, and hence do not require the existence of the input data file.

## 5 How to ... ?

### 5.1 compute stability of MHD modes

Run MARS-F as an eigenvalue solver. Set `NCASE=1`, `INCFEED=0`, `INCKIN=0`, `NSPECIES=2`. Set/change maximal number of iteration `NITMAX`.

For ideal MHD mode stability computations, set `ETA=0.0`. For resistive mode, set a finite value of `ETA`, which corresponds to the inverse of the Lundquist number at the magnetic axis. The radial profile of the plasma resistivity is controlled by `NPROFT`. To improve numerical accuracy, it is often useful to choose a proper coordinate system depending on the eigenmode structure. For example, if the mode tends to have a ballooning structure in the LFS of the torus (this is the case for most pressure driven kink or RWM), it's better to choose an equal-arc coordinate system that better resolves the LFS region than a SFL system. On the other hand, the SFL system better resolves the  $\mathbf{B} \cdot \nabla$  operator near rational surfaces (natural separation of resonant harmonics), and hence is more suitable for the TM.

Set `NV=1` if no vacuum region is included in the computations - this is equivalent to the ideal-wall boundary condition applied at the plasma surface. To include the full vacuum region, set the same value of `NV` as that defined in the CHEASE input Namelist `NV`. The ideal-wall boundary condition is then applied at the computational boundary specified by `NV`. Note that the MARS-\* input `NV` can also be used to scan the ideal wall position. In this case, the MARS-F `NV` does not have to be equal to (but should not exceed) the CHEASE input `NV`.

To include the resistive wall (e.g. for studying the RWM or computing the ac-RMP response), set `NWALL`=the number of walls (0 means that no resistive wall is included). Also set the radial position of wall(s) by `IWALL`, and the wall resistivity via the fictitious wall time array `TAUW` (see Sec. 2.7 for more details).

The most critical issue with the MARS-\* eigenvalue solver is to set the proper initial guess `TALPHA1`. This is required by the numerical algorithm (inverse vector iteration) implemented into the code. The proper choice of the initial guess requires certain *a-priori* knowledge or experience. Since the true eigenvalue of the MHD mode is generally a complex number, it is not easy, and not recommended (to avoid finding numerical spurious solutions), to start with an arbitrary initial guess in the complex plane. On the other hand, it is often possible, by varying certain physics parameter(s), to convert the case under consideration to a case where it is known that the true eigenvalue is real. For the latter case, it is recommended, unless an *a-priori* knowledge is available, to set the large value for the initial guess `TALPHA1`, say close to 1 (re-call that the eigenvalue in MARS-\* is normalised by the toroidal Alfvén frequency). If MARS-\* run does not reach convergence within the allocated number of iteration `NITMAX`, use the (un-converged) eigenvalue from MARS-\* output file `RESULT.OUT` as the new initial guess. And repeat the above procedure until the converged (real) eigenvalue is found. After finding the true (real) eigenvalue for the converted case, it is easier to trace back to the original case by gradually scanning certain physics parameter(s).

It is always a good idea to check the eigenfunctions to ensure that a physical mode is obtained. The judgement often relies on *a-priori* knowledge or experience, unfortunately, on the expected eigenstructure and eigenvalue.

## 5.2 compute mode stability with toroidal plasma flow

Follow the same setting as in Sec. 5.1. In addition, specify the angular velocity of toroidal plasma flow via the Namelist variables `NPROFR`, `ROTE` etc., as well as the rotation profile `PROFROT.IN` if `NPROFR=4`.

It is important to note that, for modes that do not rotate together with the plasma, the MHD continuum (Alfvén and sound waves) resonances with the mode occur. And these resonances do not occur exactly at the rational surfaces. A good numerical resolution of these resonances requires sufficient mesh packing near the (splitted) resonant surfaces rather than the exact rational surfaces. This can be achieved either by repetitive manual radial mesh packing (`NMESH=2`) with CHEASE, or by automatic packing (`NMESH=4` or `6`). The latter is approximate, and becomes less accurate with increasing rotation speed.

Examples where the aforementioned resonances occur include the RWM with plasma flow, the rotating plasma response to an externally applied static magnetic field perturbation.

## 5.3 study mode stability with feedback control

An important functionality of MARS-F is the capability of direct modelling of magnetic feedback control of MHD modes. There are generally two practically straightforward ways of running MARS-F to perform the feedback study (Refer to [9] for other ways of more theoretical interest). One is the direct feedback run. The other is via the computation of the plasma response model (PRM).

In the direct feedback run, both the MHD equations and an additional equation for the feedback law is jointly solved as an eigenvalue problem, with a pre-specified feedback gain. Direct feedback runs always involve three essential ingredients: (i) the active coil (actuator), (ii) the magnetic pick-up coil (sensor), and (iii) the feedback logic (controller). The active coil is specified by the number of coils `NCOIL` along the poloidal angle, the coil geometry parameters `FCCHI`, `FWCHI`, `BTCHI`, the radial position `IFEED` as well as the coil current `FEEDI` (see Sec. 5.4 for more details). The sensor coils are specified by the number of sensor positions `NSENS`, the sensor geometry `SCCHI`, `SWCHI`, and the radial positions `ISENS`. The feedback logic is specified by `INCFEED`, `GAINA`, `GAINP`, `FNUP`, `FNUD`, `IDYNAM`, and `TAUF`. Only proportional and derivative actions are considered in the feedback law. Most frequently used option is the so-called current control (`IDYNAM=0`), with either point-wise radial (`INCFEED=11`) or internal poloidal (`INCFEED=3`) sensors. Refer to Tables 9-10 for more detailed specification of the above three aspects.

For a linear control system, it is often desirable (and possible) to separate the system dynamics from the controller. Knowing the system dynamics, which can be specified by transfer functions of the open-loop response (in the frequency domain, called plasma response model, PRM), it's a separate task to design a suitable controller or perform a controller optimisation. The choice of the controller can certainly be beyond the PD-controller in this case. MARS-F can be used to obtain the PRM, via frequency scan, by computing the plasma response to prescribed current(s) in active coils. In these computations, the active and sensor coils are specified in the same way as for the direct feedback run, but there is no need to specify the feedback logic. Set the control parameter `INCFEED=4` for vacuum response (essentially the resistive wall response without plasma), and `INCFEED=8` for the plasma+wall response.

For a linear control with PD controller, The above two approaches should yield equivalent results. For the

direct feedback approach, an option is also available ([NCASE=4](#)) to study the non-linear feedback with coil voltage or current saturation, by running MARS-F as an initial value code.

## 5.4 compute vacuum fields from magnetic coils

Any external vacuum field is introduced into MARS-\* via currents. In the case of error fields, we introduce a principle of equivalent surface current (ESC) [10]. Here we consider a normal situation where the vacuum field is produced by a given set of coils with known geometry.

The vacuum field in MARS-F is computed by directly solving two equations  $\nabla \times \mathbf{b} = \mathbf{j}_{\text{coil}}$ , ( $\mu_0 = 1$ ) and  $\nabla \cdot \mathbf{b} = 0$  in the whole computational domain. It is often useful to compare the MARS-\* computed vacuum field with that produced by a Biot-Savart based code. A good agreement should be obtained for the far field (i.e. the field not very close to the coil locations). The essential part of the vacuum runs, as well as the plasma response runs described in Sec. 5.5, is the pre-specification of the input parameters for the coils in terms of MARS-\* variables. These input parameters include (i) the coil geometry, (ii) the coil current and (iii) the coil current frequency.

In MARS-\*, the coil current is always represented as a divergence-free surface current, with poloidal and toroidal components. Both components have  $\exp(in\phi)$  dependence along the toroidal angle  $\phi$ , for a given toroidal harmonic  $n$ . The toroidal component  $j_\phi^{\text{coil}}$  of the coil current density, as a function of the poloidal angle  $\chi$ , is represented by  $\delta$ -like functions (e.g. Gaussian), with the centre specified by [FCCHI](#) and the width (effectively the width of the coil's legs) specified by [BTCHI](#). The shape of these functions are specified by [KKF](#). Normally the far field is not sensitive to the details of the choice of these  $\delta$ -like functions. The poloidal component of the coil current follows the divergence-free condition for the surface current. Geometrically, this current representation corresponds to infinite number (in order to obtain a pure single  $n$  harmonic) of window-shaped coils along the toroidal angle. The location of the coils in the  $(R, Z)$ -plane is specified by the radial location of the coils [IFEED](#), the centre of the coil [FCCHI](#), and coils poloidal coverage [FWCHI](#). Therefore, one needs to convert the physical coil positions, normally specified in terms of  $(R, Z)$ -coordinates, to the aforementioned MARS-\* variables [IFEED](#), [FCCHI](#), [FWCHI](#). A Matlab script is available for this purpose.

The above specified coil current density is eventually represented by poloidal Fourier harmonics in MARS-\*. The fact that the same amount of current should flow in all legs of the window coil requires a symmetry in the choice of the band limits of the poloidal harmonics, namely [M1=-M2](#).

Since the coil current density is represented in the Fourier expansion along the poloidal angle, it is often numerically more accurate to run the code in a coordinate system with equal-arc or geometric poloidal angle to better represent the coil geometry. It's also necessary to choose such a coordinate if we shall use the same coil configuration to compute the plasma response which has a strong kink-ballooning component. On the other hand, in RMP problems, we are often interested in the amplitude of resonant harmonics as defined in a SFL system, at rational surfaces. This can be obtained via a post-processing (Matlab) script that maps the MARS-\* computed perturbed magnetic field from the equal-arc (or geometric) system to the SFL system. It is also possible to directly run the code in the equal-arc (geometric) system and output the amplitude of resonant harmonics as defined in the SFL system. An example will be shown in Sec. 6.4.



The amplitude of the  $n$ -th toroidal component of the coil current is specified by `FEEDI`, which is generally a complex number. Because in real experiments, there is always a finite number of coils along the toroidal angle, a conversion factor between the coil current with discrete coils, and the MARS-\* single- $n$  quantity `FEEDI`, need to be calculated. This conversion factor depends on how the coil currents are physically arranged, as well as on the toroidal coverage of each coil.

To perform the vacuum runs, use the option `INCFEED=4`. For the ac field, the normalised coil current frequency is specified by setting a pure imaginary number `TALPHA1`. In particular, set `TALPHA1=(0.,0.)` for the dc excitation field. If the resistive wall response to an ac vacuum field is to be computed, the parameters associated with the resistive wall, `NWALL`, `IWALL`, `TAUW`, also need to be specified.

## 5.5 compute plasma response to external fields

This often so-called RMP, and sometimes RFA, response computations share the same methodology as the PRM computations described in Sec. 5.3. The setup of the excitation coils (geometry, current and frequency) follows Sec. 5.4. Here we describe some details of physics modules that can be included into the plasma response computations. These include (i) the flow screening, (ii) the ideal versus resistive plasma response, (iii) the parallel sound wave damping model, (iv) the drift kinetic damping physics.

MARS-\* so far only includes toroidal plasma flow. In terms of fluid approximation, the code is capable of exploiting the flow screening effect in the so-called resistive-inertia regime [7]. Another theoretically interesting regime (called GGJ-regime), which occurs for finite beta plasmas at very slow rotation, can also be explored [11]. A brief discussion of the flow screening physics, as compared to the two-fluid screening due to perpendicular flow, is discussed in the Appendix B of Ref. [10].

The parallel sound wave damping model follows a multi-pole based, fluid-like theory for the proper approximation of the drift kinetic closure [12]. In MARS-\*, this term is introduced as a viscous force added to the parallel projection of the perturbed momentum equation. Physically, this model describes, to certain approximation, the ion acoustic Landau resonance induced damping along the parallel motion of thermal ions. This model is included into MARS-\* runs by setting `IVISC=1`. A numerically adjustable input coefficient `PVISC` is associated with the strength of the damping, with the nominal value of order unity. A very small (say 0.1) or vanishing value of `PVISC` excludes the parallel sound wave damping physics from the MARS-\* runs.

It is also possible to include the drift kinetic damping physics, developed in full toroidal geometry [4], into the plasma response computations. The detailed setup of the drift kinetic model is described in Sec. 5.7.

## 5.6 compute wall time

For the purpose of the RWM stability study, or the penetration of an ac RMP field through the wall, we may wish to compute the eddy current decay time of the the resistive wall in the vacuum. Normally we are interested in the slowest decaying wall time, which corresponds to the wall eigenfunction with the least stable eigenvalue, although it is also possible to compute any wall eigenfunction with MARS-\*.



Set `INCFEED=2`, `NWALL`=number of resistive walls to be included, arrays specifying the wall radius `IWALL` and the “artificial” wall time `TAUW`. The latter practically specifies the wall resistivity (cf. Sec. 2.7). Set the initial guess `TALPHA1` (negative real part and vanishing imaginary part), and run MARS-\* as an eigenvalue solver (`NCASE=1`) for the damping rate  $\gamma$  of the wall mode. The computed wall time is equal to  $-1/\gamma$ .

## 5.7 include drift kinetic effects

Set `INCKIN=1` in order to include drift kinetic effects into MARS-\* computations. There are generally two ways of including these effects: the perturbative and non-perturbative runs.

In perturbative runs (`IPERTURB`> 0), a pre-computed eigenfunction (e.g. the fluid RWM eigenfunction) is used in evaluating the kinetic terms (e.g. the perturbed drift kinetic potential energies). Also set `ATAU`=the (complex) frequency of the pre-computed eigenmode.

The non-perturbative option (`IPERTURB=0`) allows self-consistent modification of the mode eigenfunctions (and eigenvalues where applicable) by kinetic effects in MARS-\* runs. As a stability eigenvalue solver, the non-perturbative approach involves an extra outer loop to resolve the non-linearity in terms of the eigenvalue, which enters (non-linearly) into the drift kinetic integrals. The maximal number of the outer loop iterations is set by `NSWEEP`. [Recall that the upper bound of the iteration number for the “inner” loop of the inverse vector iteration in the eigenvallue solver is set by `NITMAX`.] Also set `ATAU`=(1.E+11,0.). The outer loop is based on a relaxation algorithm (combined with the Newton method). The relaxation parameter `ALTAU` can be varied to optimise the numerical convergence.

For normal runs, set `KFASTRUN=0`, `KENORM=2`, `KEFORM=2`. The parameter `NSPECIES` sets the number of particle species to be included into the kinetic runs. This is also the size of the input arrays `ISPECIES_F0`, `ESPECIES_*` and `PSPECIES_*`. The default set for the first two species are always thermal ions and electrons. The energetic particles can have more than one species (equilibrium distribution functions), starting from the particle species number 3 and so on. The arrays `ESPECIES_*` specify basic quantities of particles (see Table 11 for details). Arrays `PSPECIES_*` specify fraction of contributions from each perturbed drift kinetic energy components, with the value of 1 corresponds to the full inclusion of the corresponding kinetic energy component (see Table 13 for details).

The type of each particle species is specified by the array `ISPECIES_F0`. Type 0 corresponds to the Maxwellian equilibrium distribution of particles. Type 1 corresponds to an isotropic slowing down model. Type 2 corresponds to an (approximate) analytic model for the D-T born  $\alpha$  particles (slowing down anisotropic). Type 3 corresponds to an anisotropic NBI beam model (Gaussian distributions in particle pitch angle space), with additional input parameters `KNBI`, `ZZETA0`, and `DZETA0`. Type 4 corresponds to anisotropic ICRH induced EP model, with additional input parameters `HHTYPE4C`, `KSMTYPE4`, `S0TYPE4`, `R0TYPE4`, as well as extra input profiles `PROFR4.IN` and `PROFS4.IN`. Refer to [6] for more details on these EP types. Note that types 1,3,4 require extra input files `PROFDA.IN` and `PROFPA.IN`, specifying the fraction of the EP’s density (w.r.t. the electron density) and pressure (w.r.t. the thermal equilibrium pressure), respectively, for each EP species. These two input data specify the radial profiles after averaging over the particle energy space and over the flux surface when necessary.

The other two important parameters, associated with the drift kinetic effects, are `ALPHAP` specifying the fraction of thermal ion pressure w.r.t. the total thermal pressure, and `OMEGACIO` specifying the on-axis thermal ion gyro-frequency normalised by the Alfvén frequency.

MARS-K so far considers only the Krook collision models for thermal particles. These models are included into the code via parameters `INUTYPE`, `NPROFUI` and `NUEFFIA` (for thermal ions), `NPROFUE` and `NUEFFEA` for thermal electrons.

Finally, it is also possible to include the first order finite orbit width (FOW) correction effect into the drift kinetic computations, for both thermal and energetic particles [6]. This is achieved by setting non-vanishing values for arrays `PSPECIES_FOWP` and `PSPECIES_FOWT`, for passing and trapped particles, respectively.

## 5.8 perform quasi-linear initial value computations

The quasi-linear initial value code MARS-Q [8] is useful for toroidal modelling of the plasma rotation braking due to external 3D fields or due to MHD events. The code captures non-linear interaction between the  $n = 0$  modification on the toroidal rotation and the  $n \neq 0$  MHD perturbations. In the case of the externally applied 3D fields, the RMP field penetration process can also be modelled to certain level of physics approximation. The key physics limitations of MARS-Q model are:

- only a single  $n \neq 0$  can be modelled at a time (and hence the quasi-linear nature). This means that the code cannot be used to study the field penetration physics at large magnetic islands.
- the model so far does not include the  $n = 0$  correction to the equilibrium (except the  $n = 0$  correction to the equilibrium flow).
- the model so far does not include poloidal flow effect which, as one projection component of the perpendicular flow, may also play a screening role to the external magnetic field.

The advantages of the MARS-Q model includes

- full toroidal geometry
- a reasonably comprehensive list of momentum sink/source physics: the resonant electromagnetic ( $\mathbf{j} \times \mathbf{b}$ ) torque, the NTV torque, the Reynolds stress torque, the momentum diffusion and pinch. In particular, the drift kinetic model from MARS-K can be used to evaluate, without approximation in geometry, the NTV torque.
- a flexible adaptive time stepping scheme is envisaged, where the MHD equations are solved via an implicit time stepping scheme.

The parameters directly associated with MARS-Q runs are grouped into the Namelist input `QLIN`, see Tables 14-15. For a typical run, set `NCASE`=6. Assign values between [0,1] to multipliers `CTJXB`,

**CTNTV**, **CTREY**, indicating the fraction of the resonant EM torque, the NTV and the Reynolds torque, respectively, to be included into the toroidal momentum balance equation. Adjust the momentum diffusion model with parameters **NPROFVD**, **TCHIM0**, **TPOWVD**. If necessary, set momentum pinch model with **NPROFVP**, **TVPINCH0**.

The adaptive time stepping scheme is controlled by the input parameters **TALPHA8** ( $< 1$ ), **TDELTALIM**, **TDELTALOW**, **TDELTAMAX**, **TDELTAMIN**. When the relative change of the solution, compared to that of the previous time step, exceeds **TDELTAMAX** (below **TDELTAMIN**), the present time step is multiplied (divided) by the factor **TALPHA8**. On the other hand, an overall upper (lower) limit for the time step is set by **TDELTALIM** (**TDELTALOW**).

The criterion for terminating the simulation is set by the total number of time steps **NSWEEP** (need to set **NITMAX**=1 with **NCASE**=6). The code allows to save the full solution into a file **MARSQ.SOL** after reaching the limit of **NSWEEP**, by setting **KSOLSAVE**=1, and to continue the simulation by reading the full solution from the file, with **KSOLREAD**=1.

The key data recording the history of the time stepping are stored in the output file **TIMEEVOL.OUT**.

## 6 Examples on ...

The following examples show typical problems that MARS-\* can solve. These certainly do not cover all the capabilities of the code.

We choose an artificial but fully toroidal, up-down symmetric plasma with strong shaping, and with  $R_0=3.0\text{m}$ ,  $a=1\text{m}$ ,  $B_0=1.5\text{T}$ ,  $I_p=1.07\text{MA}$ . For various purposes, we scan the plasma pressure, via the CHEASE input parameter **CFBAL**, while fixing the total plasma current at 1.07MA. For the base case (**CFBAL**=1.0), we have an equilibrium with  $\beta_N = 1.23$ ,  $q_0 = 1.48$ ,  $q_{95} = 5.02$ ,  $q_a = 6.25$ . All the examples below study the  $n = 1$  instability or the  $n = 1$  plasma response.

These examples are stored in the folder **EXAMPLE**, which contains the following sub-folders:

- **Data**: input profiles for CHEASE and MARS-\*
- **Equi**: working directory for running CHEASE, containing sample Namelist input and output files
- **Kink**: working directory of sample MARS-F runs for computing stability of the pressure driven ideal kink mode, containing sample Namelist input and output files (Sec. 6.1)
- **RwmFluid**: working directory of sample MARS-F runs for computing stability of the pressure driven RWM, within the fluid theory (Sec. 6.2)
- **RwmKinetic**: working directory of sample MARS-F runs for computing stability of the pressure driven RWM, with the MHD-drift kinetic hybrid approach (Sec. 6.3)
- **RmpLin**: working directory of sample MARS-F runs for computing linear plasma response to external RMP fields (Sec. 6.4)
- **RmpQLin**: working directory of sample MARS-F runs for quasi-linear initial value modelling of plasma rotational damping due to the applied RMP field, as well as the associated RMP field penetration into the plasma (Sec. 6.5)
- **Tm**: working directory of sample MARS-F runs for computing the stability of the linear tearing mode (Sec. 6.6)
- **Memo**: a memo file “memo.m”, in the form of a Matlab script, recording the key elements of running the above sample cases.

### 6.1 computing stability of ideal kink mode

Here we scan the plasma pressure while fixing the total plasma current. By computing the linear growth rates of the  $n = 1$  ideal kink mode without and with an ideal wall, we extrapolate the corresponding no-wall and ideal-wall  $\beta_N$  limits, which turn out to be 3.3887 and 4.9369, respectively, for this example.

Note that by fixing the total plasma current, the  $q$ -profile is not fixed. In particular, the edge  $q$  value varies crossing the integer number 9, resulting in a change of the stability behaviour for the ideal-wall

ideal kink mode. Also note that this test plasma is rather stable against the  $n = 1$  pressure driven ideal kink instability - the computed  $\beta_N$  limits are rather high compared to typical plasmas from experiments.

## 6.2 computing stability of fluid RWM

Here we give three examples for computing, respectively, (i) the wall time which corresponds to the slowest decay time of the  $n = 1$  wall eigenmode in vacuum (without plasma); (ii) the stability of the RWM without plasma flow; and (iii) the stability of the RWM with plasma flow. In the latter example, a small rotation frequency of  $\Omega_0 = 10^{-3}\omega_A$  is assumed. Couple of key notes on computing the RWM stability with flow

- It is always desirable to scan the mode's eigenvalue starting from slow rotation - it is practically difficult to make a proper initial guess for the mode's complex eigenvalue starting with an arbitrary flow speed.
- With increasing equilibrium flow speed, continuum MHD waves induced resonant splitting becomes more prominent. In order to properly resolve these resonant surfaces (which occur on both sides the original rational surfaces), re-packing of the radial mesh in the plasma region is often required (see Sec. 5.2).

These examples only show the stability of the RWM with  $C_\beta \equiv (\beta_N - \beta_N^{\text{nw}})/(\beta_N^{\text{iw}} - \beta_N^{\text{nw}}) = 0.5$ . Similar computations can be performed for other  $C_\beta$  values in order to map out the instability of the whole RWM-regime. The present example assumes a negligible parallel sound wave damping effect ([PVISC=0.1](#)). Within the fluid approximation, it is also desirable to consider how the mode's stability is affected by this damping physics.

## 6.3 computing RWM stability with kinetic effects

Here we again consider an equilibrium with  $C_\beta = 0.5$ , and assume a slow plasma flow of  $\Omega_0 = 10^{-3}\omega_A$ . In terms of the kinetic aspects, the sample run assumes: (i) the non-perturbative approach, (ii) the inclusion of thermal particle kinetic effects only ([NSPECIES=2](#)), and (iii) the precessional drift resonance effects only. The other kinetic effects can also be included by a proper setup of the Namelist input parameters (see Sec. 5.7).

For kinetic computations, it is important to set the proper value for the Namelist variable [OMEGACIO](#) - most of the kinetic effects scale with this parameter.

## 6.4 computing plasma response to RMP fields

The RMP response computations require certain initial steps before the actual runs can be performed. This is mostly related to the setup of the coil geometry and currents, see Sec. 5.4. In the shown example, these input data are simply assumed as known.

In order to allow the code to output the field spectrum (for the resonant harmonics) in the SFL coordinate system, while running the code in a non-SFL system, certain preparatory actions are envisaged, as shown in this example (refer to EXAMPLE/Memo/memo.m for more details).

Both vacuum field and the plasma response field are computed here. The vacuum results can be benchmarked against Biot-Savart integral based codes. The plasma response results depend on the physics models that are included into the computations. Within the fluid version, typically three physics can be explored: the flow screening, the plasma resistivity, and the parallel sound wave damping. The sample run here assumes  $\Omega_0 = 10^{-3}\omega_A$ ,  $S(s=0) = 10^7$  and **PVISC**=1.5. The inclusion of the drift kinetic effects for the plasma response computations follows similar setups as that for the non-perturbative computations of the RWM stability.

## 6.5 modelling RMP induced rotation braking

The quasi-linear initial value modelling (MARS-Q run) is most time-consuming, since a typical run requires time stepping with an implicit scheme (to avoid numerical instabilities). The sample run here is terminated after 20 time steps (**NSWEEP**=20), although a proper run (till saturation if there is such a solution) may require several hundreds (or more) time steps.

The input adaptivity parameters as described in Sec. 5.8 may require certain tuning from one case to another, in order to ensure efficient time stepping while without violating the evolution physics of the rotation braking and the field penetration process.

## 6.6 computing growth rate of TM

The examples here show typical computations of the TM stability within the single fluid theory. Two cases are shown. Case 1 scans the Lundquist number  $S$  for a pressure-less equilibrium (**CFBAL**=0), showing the correct scaling of  $S^{-3/5}$  for the TM growth rate, as  $S$  becomes large. Case 2 scans the equilibrium pressure **CFBAL** while fixing the Lundquist number at  $S = 10^6$ . At sufficiently large pressure ( $\beta_N \simeq 0.14$  in this example), the eigenvalue of the TM becomes complex (in a static plasma) due to the toroidal favourable average curvature stabilisation effect.

Examination of the mode's eigenfunction shows that the TM occurs at multiple rational surfaces ( $q = 2, 3, 4, 5$ ), with the  $m/n = 2/1$  component being the dominant one.

Attention should to paid to the radial mesh packing near rational surfaces when computing the TM stability with large  $S$  values. With sufficient mesh packing, MARS-\* can compute the TM stability with the  $S$  values as high as  $10^{10}$ .

The computations of the TM stability can also be extended by including other physics modules.

## 7 FAQ

This Section was compiled by David Ryan.

Q: What “MARS” stands for?

A: MAgnetohydrodynamic Resistive Spectrum.

Q: What are the hardware requirements for running MARS-\*?

A: 32-64GB RAM recommended, but not essential. Code parallelises over many processors, so multiple processors recommended, but not essential.

Q: I’m having trouble compiling MARS-F on my machine, are there any examples I can use?

A: Included in the source code distribution is a collection of makefiles, which have been used to compile MARS-F on a wide variety of machine architectures. Because of extensive use of parallelism in the code, it is advised to use a parallel compiler wrapper such as mpif90.

Q: How do I generate input files for MARS-\*?

A: MARS-\* requires a plasma equilibrium found by Grad-Shafranov equilibrium solver CHEASE.

Q: How do I use CHEASE?

A: CHEASE reads input from a text file called 'datain', which contains a list of all the variables required by CHEASE. A sample 'datain' file can be found in the MARS distribution. Also in the distribution is an example of a shell script (Rchease\_EXAMPLE\_base), which creates the 'datain' file, and then uses it to launch CHEASE. CHEASE outputs files named OUTVMAR and OUTRMAR, which are the input files for MARS-F.

Q: How do I run MARS-\*?

A: MARS reads input from OUTVMAR and OUTRMAR (which contain the equilibrium data), and RUN.IN, which contains all the other variables which MARS-\* requires. A sample 'RUN.IN' file is included in the distribution, as well as a shell script which generates it, and launches and times MARS-\*.

Q: Running MARS-\* using the example script produces the following error: forrtl: severe (24): end-of-file during read, unit 31, file /home/dryan/MarsQ/MARSQ.SOL

A: The script has attempted to read a file which doesn't exist. Set input parameter KSOLREAD to 0 to avoid this.

Q: What does MARS-\* output?

A: Basic output is written to the following files:

- RMZM\_F.OUT : equilibrium magnetic geometry
- BPLASMA.OUT : computed perturbed B-field
- VPLASMA.OUT : perturbed plasma velocity
- XPLASMA.OUT : plasma displacement
- JPLASMA.OUT : perturbed plasma current
- PPLASMA.OUT : perturbed plasma pressure
- RESULT.OUT : eigenvalues of modes (ie, the spectrum)



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